

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only.

TORQUITE TABLETS 25 MG

TORQUITE TABLETS 300 MG

(QUETIAPINE FUMARATE TABLETS 25 MG AND 300 MG)

BRAND OR PRODUCT NAME

TORQUITE TABLETS 25 MG

TORQUITE TABLETS 300 MG

NAME AND STRENGTH OF ACTIVE INGREDIENT

TORQUITE TABLETS 25 MG

TORQUITE TABLETS 300 MG

Each film coated tablet contains: Quetiapine Fumarate Ph.Eur.

equivalent to Quetiapine 25 mg

Colours: Yellow oxide of Iron, Red Oxide of Iron and Titanium Dioxide

TORQUITE TABLETS 300 MG

Each film coated tablet contains: Quetiapine Fumarate Ph.Eur.

equivalent to Quetiapine 300 mg

Colour: Titanium Dioxide

PRODUCT DESCRIPTION

TORQUITE TABLETS 25 MG

Brownish red coloured, round, biconvex, film coated tablets plain on both sides.

TORQUITE TABLETS 300 MG

White to off white, capsule shaped, biconvex, film coated tablets with break line on both sides.

DOSAGE FORM

Film coated tablet

Pharmacodynamic Properties

Pharmacotherapeutic group: Antipsychotics

Mechanism of action

Quetiapine is an atypical antipsychotic agent. Quetiapine and the active human plasma metabolite, norquetiapine interact with a broad range of neurotransmitter receptors. Quetiapine and norquetiapine exhibit affinity for brain serotonin (5HT₂) and dopamine D₁ and D₂-receptors. It is this combination of receptor antagonism with a higher selectivity for 5HT₂ relative to D₂-receptors, which is believed to contribute to the clinical antipsychotic properties and low extrapyramidal side effect (EPS). Quetiapine has no affinity for the norepinephrine transporter (NET) and low affinity for the serotonin 5HT_{1A} receptor, whereas norquetiapine has high affinity for both. Inhibition of NET and partial agonist action at 5HT_{1A} sites by norquetiapine may contribute to quetiapine's therapeutic efficacy as an antidepressant. Quetiapine and norquetiapine have high affinity at histaminergic and adrenergic alpha 1 receptors and moderate affinity at adrenergic alpha2 receptors. Quetiapine also has low or no affinity for muscarinic receptors, while norquetiapine has moderate to high affinity for several muscarinic receptor subtypes.

Pharmacodynamic effects:

Quetiapine is active in tests for antipsychotic activity, such as conditioned avoidance. It also blocks the action of dopamine agonists, measured either behaviourally or electro physiologically and elevates dopamine metabolite concentrations, a neurochemical index of D2-receptor blockade.

Pharmacokinetic properties

Absorption

Quetiapine is well absorbed and extensively metabolised following oral administration. The bioavailability of quetiapine is not significantly affected by administration with food. Steady-state peak molar concentrations of the active metabolite norquetiapine are 35% of that observed for quetiapine. The pharmacokinetics of quetiapine and norquetiapine are linear across the approved dosing range.

Distribution

Quetiapine is approximately 83% bound to plasma proteins.

Biotransformation

Quetiapine is extensively metabolised by the liver, with parent compound accounting for less than 5% of the unchanged drug-related material in the urine or faeces, following the administration of radiolabelled quetiapine. In vitro investigations established that CYP3A4 is the primary enzyme responsible for cytochrome P450 mediated metabolism of quetiapine. Norquetiapine is primarily formed and eliminated via CYP3A4. Approximately 73% of the radiolabelled quetiapine excreted in the urine and 21% in the faeces. Quetiapine and several of its metabolites (including norquetiapine) were found to be weak inhibitors of human cytochrome P450 1A2, 2C9, 2C19, 2D6 and 3A4 activities in vivo. In vitro CYP inhibition is observed only at concentrations approximately 5 to 50 fold higher than those observed at a dose range of 300 to 800 mg/day in humans. Based on these in vitro results, it is unlikely that co-administration of quetiapine with other drugs will result in clinically significant drug inhibition of cytochrome P450 mediated metabolism of the other drug. From animal studies it appears that quetiapine can induce cytochrome P450 enzymes. In a specific interaction study in psychiatric patients, however, no increase in the cytochrome P450 activity was found after administration of quetiapine.

Elimination

The elimination half-lives of quetiapine and norquetiapine are approximately 7 and 12 hours, respectively. The average molar dose fraction of free quetiapine and the active human plasma metabolite norquetiapine is <5% excreted in the urine.

Special populations

Gender

The kinetics of quetiapine do not differ between men and women.

Elderly

The mean clearance of quetiapine in the elderly is approximately 30 to 50% lower than that seen in adults aged 18 to 65 years.

Patients with renal impairment

The mean plasma clearance of quetiapine was reduced by approximately 25% in subjects with severe renal impairment (creatinine clearance less than 30 ml/min/1.73m²), but the individual clearance values are within the range for normal subjects.

Patients with hepatic impairment

The mean quetiapine plasma clearance decreases with approx. 25% in persons with known hepatic impairment (stable alcohol cirrhosis). As quetiapine is extensively metabolised by the liver, elevated plasma levels are expected in the population with hepatic impairment. Dose adjustments may be necessary in these patients.

Paediatric population

Pharmacokinetic data were sampled in 9 children aged 10-12 years old and 12 adolescents, who were on steady-state treatment with 400 mg quetiapine twice daily. At steady-state, the dose-normalised plasma levels of the parent compound, quetiapine, in children and adolescents (10-17 years) were similar to those observed in adults, though Cmax in children was at the higher end of the range observed in adults. The AUC and Cmax for the active metabolite, norquetiapine, were higher, approximately 62% and 49% in children (10-12 years), respectively and 28% and 44% in adolescents (13-17 years), respectively, compared to adults.

Indications

- Treatment of schizophrenia.
- Treatment of acute manic episodes associated with bipolar I disorder, as either mono therapy or adjunct to lithium or divalproex.
- Treatment of depressive episodes associated with bipolar disorder.
- Maintenance treatment of bipolar I disorder as adjunct therapy to lithium or divalproex.**

Special Considerations in Treating Pediatric Schizophrenia and Bipolar I Disorder

Pediatric schizophrenia and bipolar I disorder are serious mental disorders, however, diagnosis can be challenging. For pediatric schizophrenia, symptom profiles can be variable, and for bipolar I disorder, patients may have variable patterns of periodicity of manic and depressive episodes. It is recommended that medication therapy for pediatric schizophrenia and bipolar I disorder be initiated only after a thorough diagnostic evaluation has been performed and careful consideration given to the risks associated with medication treatment. Medication treatment for both pediatric schizophrenia and bipolar I disorder is indicated as part of a total treatment program that often includes psychological, educational and social interventions.

- Prevention of recurrence in patients with bipolar disorder, in patients where manic, or depressive episode has responded to quetiapine treatment

Posology and method of administration

Mode of Administration

Oral

Adults

Quetiapine should be administered twice daily, with or without food. Quetiapine should generally be administered with an initial dose of 25 mg BID, with increases in increments of 25-50 mg BID or TID on the second and third day, as tolerated, to a target dose range of 300 to 400 mg daily by the fourth day, given BID or TID. Further dosage adjustments, if indicated, should generally occur at intervals of not less than 2 days, as steady state for Quetiapine would not be achieved for approximately 1, 2 days in the typical patient when dosage adjustments are necessary. Dose increments/decrements of 25-50 mg BID are recommended.

Adolescents (12-17 years)

Quetiapine should be administered twice daily. However, based on response and tolerability Quetiapine may be administered three times daily where needed. The total daily dose for the initial five days of therapy is 50 mg (Day 1), 100 mg (Day 2), 200 mg (Day 3), 300 mg (Day 4) and 400 mg (Day 5). After Day 5, the dose should be adjusted within the recommended dose range of 400 mg/day to 800 mg/day based on response and tolerability. Dosage adjustments should be in increments of no greater than 100 mg/day.

Maintenance Treatment

While there is no body of evidence available to answer the question of how long the patient treated with Quetiapine should remain on it, the effectiveness of maintenance treatment is well established for many other antipsychotic drugs. It is recommended that responding patients be continued on Quetiapine, but at the lowest dose needed to maintain remission. Patients should be periodically reassessed to determine the need for maintenance treatment.

Reinitiation of Treatment in Patients Previously Discontinued

Although there are no data to specifically address re initiation of treatment, it is recommended that when restarting patients who have had an interval of less than one week off Quetiapine, titration of Quetiapine is not required and the maintenance dose may be reinitiated. When restarting therapy of patients who have been off Quetiapine for more than one week, the initial titration schedule should be followed.

Acute manic episodes associated with bipolar I disorder:

Adults

Quetiapine should be administered twice daily, with or without food. As monotherapy or as adjunct therapy to mood stabilizers (lithium or divalproex), the total daily dose for the first four days of therapy is 100 mg (Day 1), 200 mg (Day 2), 300 mg (Day 3) and 400 mg (Day 4). Further dosage adjustments up to 800 mg per day by Day 6 should be in increments of no greater than 200 mg per day. The dose may be adjusted depending on clinical response and tolerability of the individual patient, within the range of 200 to 800 mg per day. The usual effective dose is in the range of 400 to 800 mg per day.

Children and Adolescents (10 to 17 years)

Quetiapine should be administered twice daily. However, based on response and tolerability.

Quetiapine may be administered three times daily where needed. The total daily dose for the initial five days of therapy is 50 mg (Day 1), 100 mg (Day 2), 200 mg (Day 3), 300 mg (Day 4) and 400 mg (Day 5). After day 5, the dose should be adjusted within the recommended dose range of 400 to 600 mg/day based on response and tolerability. Dosage adjustments should be in increments of no greater than 100 mg/day.

Depressive episodes associated with bipolar disorder:

Quetiapine should be administered once daily at bedtime, with or without food. Quetiapine should be titrated as follows: 50 mg (Day 1), 100 mg (Day 2), 200 mg (Day 3) and 300 mg (Day 4). Quetiapine can be titrated to 400 mg on Day 5 and up to 600 mg by Day 8.

Maintenance treatment of bipolar I disorder as adjunct therapy to lithium or divalproex.

Adults

Generally, in the maintenance phase, patients continued on the same dose on which they were stabilized during the stabilization phase.

Children and Adolescents (10 to 17 years)

It is generally recommended that responding patients be continued beyond the acute response, but at the lowest dose needed to maintain remission. Patients should be periodically reassessed to determine the need for maintenance treatment.

Preventing recurrence in bipolar disorder

For preventing recurrence of manic, mixed or depressive episodes in bipolar disorder, patients who have responded to quetiapine for acute treatment of bipolar disorder should continue therapy at the same dose. The dose may be adjusted depending on clinical response and tolerability of the individual patient, within the range of 300 to 800 mg/day administered twice daily. It is important that the lowest effective dose is used for maintenance therapy.

Elderly

As with other antipsychotics, Quetiapine should be used with caution in the elderly, especially during the initial dosing period. The rate of dose titration may need to be slower, and the daily therapeutic dose lower, than that used in younger patients, depending on the clinical response and tolerability of the individual patient.

Pediatric Use

Increases in systolic and diastolic blood pressure occurred in children and adolescents and did not occur in adults. Orthostatic hypotension occurred more frequently in adults (4-7%) compared to children and adolescents (<1%).

Schizophrenia

Safety and effectiveness of Quetiapine in pediatric patients less than 13 years of age with schizophrenia have not been established.

Maintenance

The safety and effectiveness of Quetiapine in the maintenance treatment of bipolar disorder has not been established in pediatric patients less than 18 years of age. The safety and effectiveness of Quetiapine in the maintenance treatment of schizophrenia has not been established in any patient population, including pediatric patients.

Bipolar Mania

Bipolar Depression

Safety and effectiveness of Quetiapine in pediatric patients less than 18 years of age with bipolar depression have not been established. Some differences in the pharmacokinetics of quetiapine were noted between children/adolescents (10 to 17 years of age) and adults. When adjusted for weight, the AUC and Cmax of quetiapine were 41% and 39% lower, respectively, in children and adolescents, compared to adults. The pharmacokinetics of the active metabolite, norquetiapine, was similar between children/adolescents and adults after adjusting for weight.

Renal impairment

Dosage adjustment is not necessary in patients with renal impairment.

Hepatic impairment

Quetiapine is extensively metabolised by the liver. Therefore, Quetiapine should be used with caution in patients with known hepatic impairment especially during the initial dosing period. Patients with known hepatic impairment should be started with 25 mg/day. The dosage should be increased daily with Increments of 25-50 mg/day until an effective dosage, depending on the clinical response and tolerability of the individual patient, is achievable.

Contraindications

Hypersensitivity to the active substance or to any of the excipients of this product. An concomitant administration of cytochrome P450 3A4 inhibitors, such as HIV-protease inhibitors, azole-antifungal agents, erythromycin, clarithromycin and nefazodone, is contraindicated.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

As quetiapine has several indications, the safety profile should be considered with respect to the individual patient's diagnosis and the dose being administered.

Paediatric population

Quetiapine is not recommended for use in children and adolescents below 18 years of age, due to a lack of data to support use in this age group. Certain adverse events occurred at a higher frequency in children and adolescents compared to adults (increased appetite, elevations in serum prolactin, vomiting, rhinitis and syncope) or may have different implications for children and adolescents (extrapyramidal symptoms and irritability) and one was identified that has not been previously seen in adult studies (increases in blood pressure). Changes in thyroid function tests have also been observed in children and adolescents.

Furthermore, the long-term safety implications of treatment with quetiapine on growth and maturation have not been studied beyond 26 weeks. Long-term implications on cognitive and behavioral development are not known.

Suicide/suicidal thoughts or clinical worsening

Depression in bipolar disorder is associated with an increased risk of suicidal thoughts, self-harm and suicide (suicide-related events). This risk persists until significant remission occurs. As improvement may not occur during the first few weeks or more of treatment, patients should be closely monitored until such improvement occurs. It is general clinical experience that the risk of suicide may increase in the early stages of recovery.

In addition, physicians should consider the potential risk of suicide-related events after abrupt cessation of quetiapine treatment, due to the known risk factors for the disease being treated. Other psychiatric conditions for which quetiapine is prescribed can also be associated with an increased risk of suicide related events. In addition, these conditions may be co-morbid with major depressive episodes. The same precautions observed when treating patients with major depressive episodes should therefore be observed when treating patients with other psychiatric disorders. Patients with a history of suicide related events, or those exhibiting a significant degree of suicidal ideation prior to commencement of treatment are known to be at greater risk of suicidal thoughts or suicide attempts, and should receive careful monitoring during treatment. Close supervision of patients, in particular those at high risk should accompany drug therapy especially in early treatment and following dose changes. Patients (and caregivers of patients) should be alerted about the need to monitor for any clinical worsening, suicidal behaviour or thoughts and unusual changes in behaviour and to seek medical advice immediately if these symptoms persist.

Metabolic Risk

Given the observed risk for worsening of their metabolic profile, including changes in weight, blood glucose (see hyperglycaemia) and lipids, which was seen in clinical studies, patients' metabolic parameters should be assessed at the time of treatment initiation and changes in these parameters should be regularly controlled for during the course of treatment. Worsening in these parameters should be managed as clinically appropriate.

The pharmacokinetics of quetiapine were not significantly altered by co-administration of the antidepressants imipramine (a known CYP 2D6 inhibitor) or fluoxetine (a known CYP 3A4 and CYP 2D6 inhibitor).

The pharmacokinetics of quetiapine were not significantly altered by co-administration of the antipsychotics risperidone or haloperidol. Concomitant use of quetiapine and thioridazine caused an increased clearance of quetiapine with approx. 70%.

The pharmacokinetics of quetiapine were not altered following co-administration with cimetidine. The pharmacokinetics of lithium were not altered when co-administered with quetiapine.

The pharmacokinetics of sodium valproate and quetiapine were not altered to a clinically relevant extent when co-administered. A retrospective study of children and adolescents who received valproate, quetiapine, or both, found a higher incidence of leucopenia and neutropenia in the combination group versus the monotherapy groups.

Formal interaction studies with commonly used cardiovascular medicinal products have not been performed. Caution should be exercised when quetiapine is used concomitantly with medicinal products known to cause electrolyte imbalance or to increase QT interval. There have been reports of false positive results in enzyme immunoassays for methadone and tricyclic antidepressants in patients who have taken quetiapine. Confirmation of questionable immunoassay screening results by an appropriate chromatographic technique is recommended.

Pregnancy

The moderate amount of published data from exposed pregnancies (i.e. between 300-1000 pregnancy outcomes), including individual reports and some observational studies do not suggest an increased risk of malformations due to treatment. However, based on all available data, a definite conclusion cannot be drawn. Animal studies have shown reproductive toxicity. Therefore, quetiapine should only be used during pregnancy if the benefits justify the potential risks.

Third trimester Neonates exposed to antipsychotics during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, tremor, extrapyramidal, respiratory distress, or feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalisation. Torquite should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

Breast-feeding Based on very limited data from published reports on quetiapine excretion into human breast milk, excretion of quetiapine through breast milk does appear to be inconsistent. Due to lack of robust data, a decision must be made whether to discontinue breast-feeding or to discontinue quetiapine therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

The effects of quetiapine on human fertility have not been assessed. Effects related to elevated prolactin levels were seen in rats, although these are not directly relevant to humans.

Effect on ability to drive and use machines Given its primary central nervous system effects, quetiapine may interfere with activities requiring mental alertness. Therefore, patients should be advised not to drive or operate machinery, until individual susceptibility to this is known.

UNDESIRABLE EFFECTS

Summary of the safety profile

The most commonly reported Adverse Drug Reactions (ADRs) with quetiapine are somnolence, dizziness, headache, dry mouth, withdrawal (discontinuation) symptoms, elevations in serum triglyceride levels, elevations in total cholesterol (predominantly LDL cholesterol), decreases in HDL cholesterol, weight gain,

decreased haemoglobin and extrapyramidal symptoms. Tabulated list of adverse reactions The incidences of ADRs associated with quetiapine therapy, are tabulated below (Table 1) according to the format recommended by the Council for International Organizations of Medical Sciences (CIOMS II Working Group 1995).

Table 1: ADRs associated with quetiapine therapy

The following frequency estimates are used in assessing adverse events:

Adverse Drug Reactions (ADRs) by System Organ Class and Frequency

MedDRA System Organ Class Frequency ADRs

Blood and lymphatic system disorders Very common Common

Decreased haemoglobin Leucopenia, Decreased neutrophil count, Eosinophils increased

Uncommon Neutropenia, Thrombocytopenia, Anaemia, Platelet count decreased

Rare Agranulocytosis

Immune system disorders Uncommon Hypersensitivity (including allergic skin reactions)

Very rare Anaphylactic reaction

Endocrine disorders Common Hypoprolactinaemia, Decreases in total T₄, Decreases in free T₄, Decreases in total T₃ increases in TSH

Uncommon Decreases in free T₃, hypothyroidism

Very rare Inappropriate antidiuretic hormone secretion

Metabolism and nutrition disorders Very common Elevations in serum triglyceride levels, Elevations in total cholesterol (predominantly LDL cholesterol), Decreases in HDL cholesterol, Weight gain

Common Increased appetite, Blood glucose increased to hyperglycaemic levels

Uncommon Hyponatraemia, Diabetes mellitus, Exacerbation of pre-existing diabetes

Rare Metabolic syndrome

Psychiatric disorders Common Abnormal dreams and nightmares, Suicidal ideation and suicidal behaviour

Rare Somnambulism and related reactions such as sleep talking and sleep related eating disorder

Nervous system disorders Very common Dizziness, Somnolence, Headache, Extrapyramidal symptoms

Common Dysarthria

Uncommon Seizure, Restless legs syndrome, Tardive dyskinesia, Syncope

Eye disorders Common Vision blurred

Cardiac disorders Uncommon Tachycardia, Palpitations

Uncommon QT prolongation, Bradycardia

Vascular disorders Common Orthostatic hypotension

Rare Venous thromboembolism

Respiratory, thoracic and mediastinal disorders Common Dyspnoea, Sleep apnoea

Atypical antipsychotic drugs, such as Quetiapine, have been associated with cases of sleep apnoea, with or without concomitant weight gain. In patients who have a history of or are at risk for sleep apnoea, Torquite Tablet should be prescribed with caution.

Uncommon Rhinitis

Very common Dry mouth

Common Constipation, Dyspepsia, Vomiting

Uncommon Dysphagia

Rare Pancreatitis, Intestinal obstruction/ileus

Hepatobiliary disorders Common Elevations in serum alanine aminotransferase (ALT), Elevations in gamma-GT levels

Uncommon Elevations in serum transaminase (AST) aspartate

Rare Jaundice, Hepatitis

Very rare Angioedema, Stevens-Johnson syndrome

Not known Toxic epidermal necrolysis, Erythema multiforme

Musculoskeletal and connective tissue disorders Very rare Rhabdomyolysis

Renal and urinary disorders Uncommon Urinary retention

Not known Drug withdrawal syndrome neonatal

Reproductive system and breast disorders Uncommon Sexual dysfunction

Rare Priapism, Galactorrhoea, breast swelling, Menstrual disorder

General disorders and administration site conditions Very common Withdrawal symptoms (discontinuation)

Common Mild asthenia, Peripheral Oedema irritability, Pyrexia

Rare Neuroleptic malignant syndrome, hyperthermia

Investigations Rare Elevations in blood creatine phosphokinase

Paediatric population The same ADRs described above for adults should be considered for children and adolescents. The following table (Table 2) summarises ADRs that occur in a higher frequency category in children and adolescent patients (10-17 years of age) than in the adult population or ADRs that have not been identified in the adult population.

Table 2: ADRs in children and adolescents associated with quetiapine therapy that occur in a higher frequency than in adults, or not identified in the adult population

The following frequency estimates are used in assessing adverse events:

Adverse Drug Reactions (ADRs) by System Organ Class and Frequency

MedDRA System Organ Class Frequency ADRs

Endocrine disorders Very common Elevations in prolactin

Metabolism and nutrition disorders Very common Increased appetite

Nervous system disorders Very common Extrapyramidal

Vascular disorders Very common Increases pressure

Respiratory, thoracic and mediastinal disorders Common Rhinitis

Gastrointestinal disorders Very common Vomiting

General disorders and administration site conditions Common Irritability

Overdose

Symptoms In general, reported signs and symptoms were those resulting from an exaggeration of the active substance's known pharmacological effects, i.e., drowsiness and sedation, tachycardia, hypotension and anti-cholinergic effects.

Overdose could lead to QT-prolongation, seizures, status epilepticus, rhabdomyolysis, respiratory depression, urinary retention, confusion, delirium, and/or agitation, coma and death. Patients with pre-existing severe cardiovascular disease may be at an increased risk of the effects of overdose.

Management There is no specific antidote to quetiapine. In cases of severe signs, the possibility of multi-drug involvement should be considered, and intensive care procedures are recommended, including establishing and maintaining a patent airway, ensuring adequate oxygenation and ventilation, and monitoring and support of the cardiovascular system.

Based on public literature, patients with delirium and agitation and a clear anticholinergic syndrome may be treated with physostigmine, 1-2 mg (under continuous ECG monitoring). This is not recommended as standard treatment, because of potential negative effect of physostigmine on cardiac conduction.